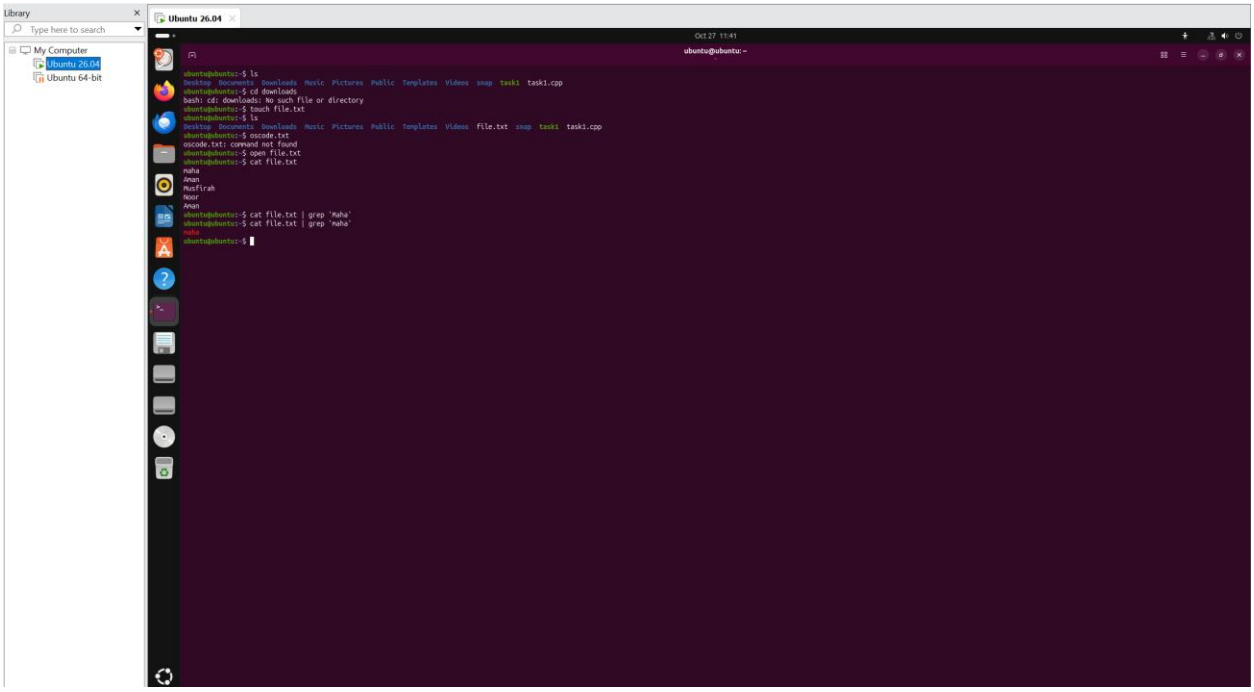
OPERATING SYSTEMS LAB ASSIGNMENT

Name: Aman SHEHZADZA

Roll No: 097

Task 1: Basic Linux Terminal Commands

Explanation: This task demonstrates basic Linux terminal commands in Ubuntu. The ls command lists the files and folders in the current directory. The cd downloads command failed with "No such file or directory" because Linux is case-sensitive and the folder is named Downloads. The touch command created a new empty file named file.txt, and cat displayed its contents (a list of names). Finally, cat was combined with grep using a pipe (|) to search for a word: searching for 'Maha' returned nothing because grep is case-sensitive, while searching for 'maha' found and highlighted the match.

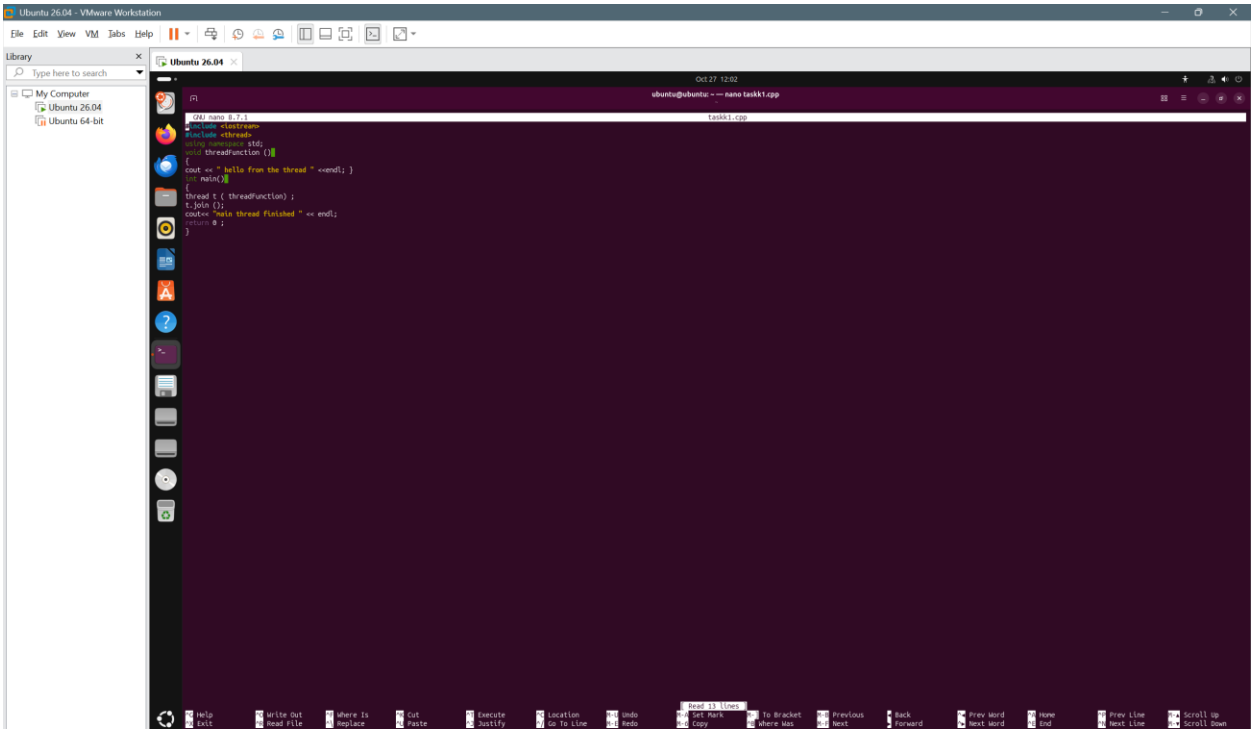


Task 2: Creating a Simple Thread

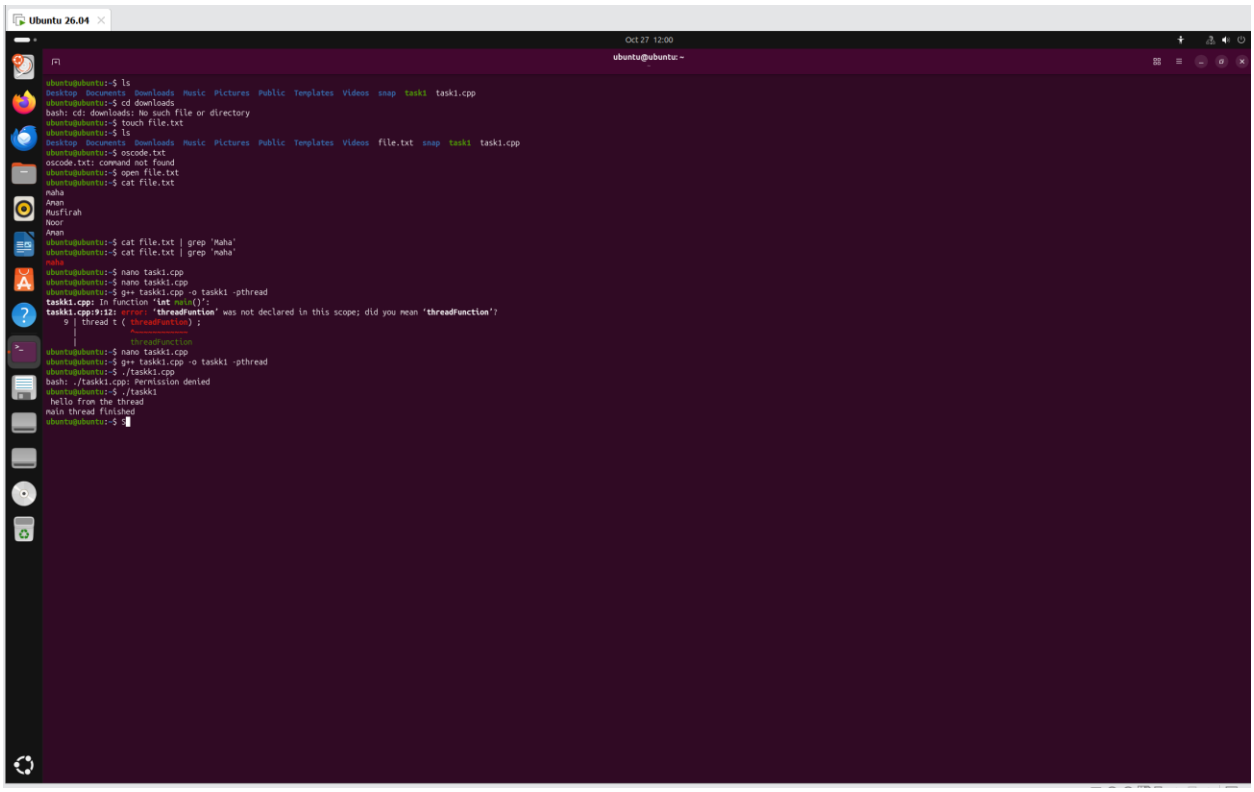
Explanation: This program shows how to create a single thread in C++ using the <thread> library. The function threadFunction() prints "hello from the thread". In main(), a thread t is created to run this function, and t.join() makes the main thread wait until the new thread finishes. Only after that does main print "main thread finished", so the output always appears in this order.

The program was compiled with g++ taskk1.cpp -o taskk1 -pthread (the -pthread flag links the threading library). The first compile attempt gave an error because the function name was misspelled; after correcting it, the program compiled and ran successfully with ./taskk1.

Code:



Output :

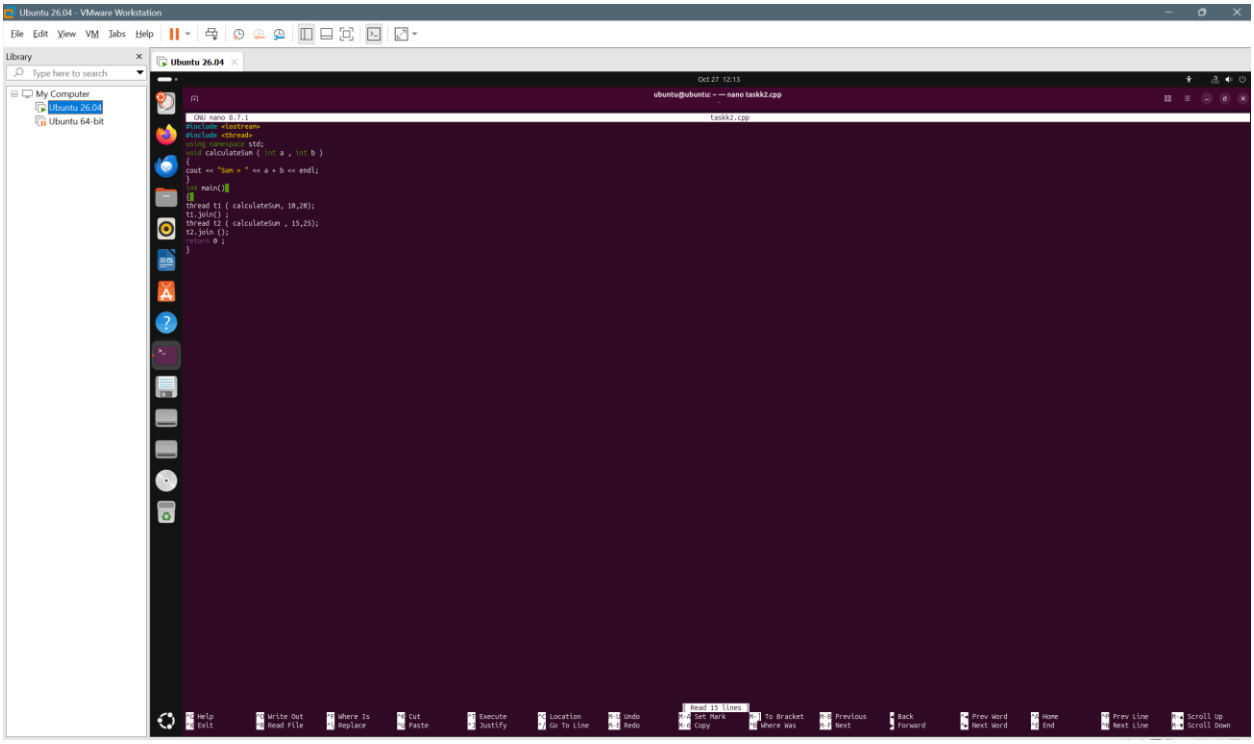

Ubuntu 26.04
Oct 27 12:00
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap task1 task1.cpp
ubuntu@ubuntu:~$ cd downloads
bash: cd: downloads: No such file or directory
ubuntu@ubuntu:~$ touch file.txt
ubuntu@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos file.txt snap task1 task1.cpp
ubuntu@ubuntu:~$ oscode.txt
oscode.txt: command not found
ubuntu@ubuntu:~$ open file.txt
ubuntu@ubuntu:~$ cat file.txt
maha
Anan
Musfirah
Noor
Anan
ubuntu@ubuntu:~$ cat file.txt | grep 'Maha'
ubuntu@ubuntu:~$ cat file.txt | grep 'maha'
maha
ubuntu@ubuntu:~$ nano task1.cpp
ubuntu@ubuntu:~$ nano task1.cpp
ubuntu@ubuntu:~$ g++ task1.cpp -o taskk1 -pthread
task1.cpp: In function 'int main()':
task1.cpp:9:12: error: 'threadFuntion' was not declared in this scope; did you mean 'threadFunction'?
9 | thread t (threadFuntion) ;
| ^~~~~~~~~~~~
| threadFunction
ubuntu@ubuntu:~$ nano task1.cpp
ubuntu@ubuntu:~$ g++ task1.cpp -o taskk1 -pthread
ubuntu@ubuntu:~$ ./task1.cpp
bash: ./task1.cpp: Permission denied
ubuntu@ubuntu:~$ ./taskk1
hello from the thread
main thread finished
ubuntu@ubuntu:~$

Task 3: Passing Arguments to Threads (Sum Calculation)

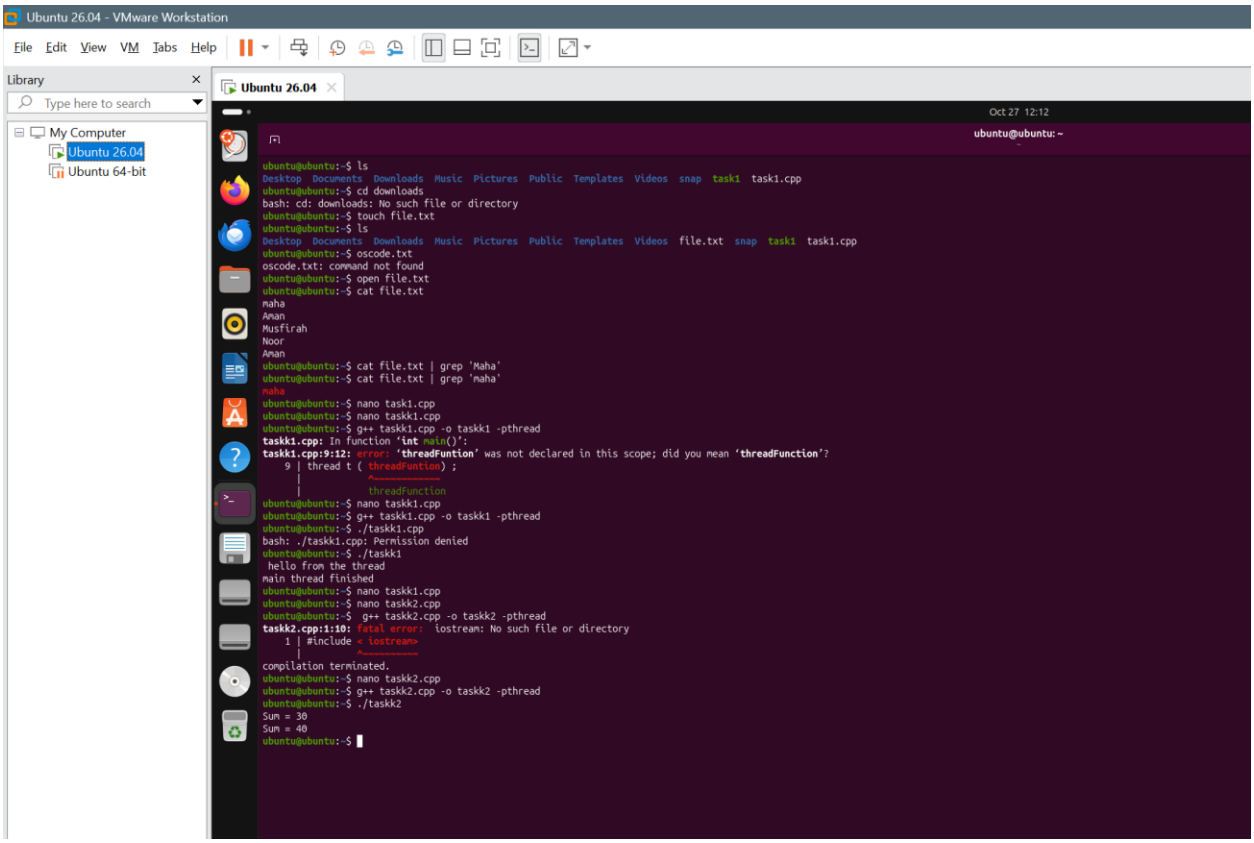
Explanation: This program shows how to pass arguments to a thread. The function calculateSum(int a, int b) prints the sum of its two parameters. Thread t1 runs it with the values 10 and 20, and thread t2 runs it with 15 and 25. Each thread is joined before the next one starts, so the results are printed in order: Sum = 30 and Sum = 40.

The first compile attempt failed because of a typo in the #include <iostream> line (an extra space). After fixing it, the program compiled with -pthread and produced the expected output.

Code :



Output:

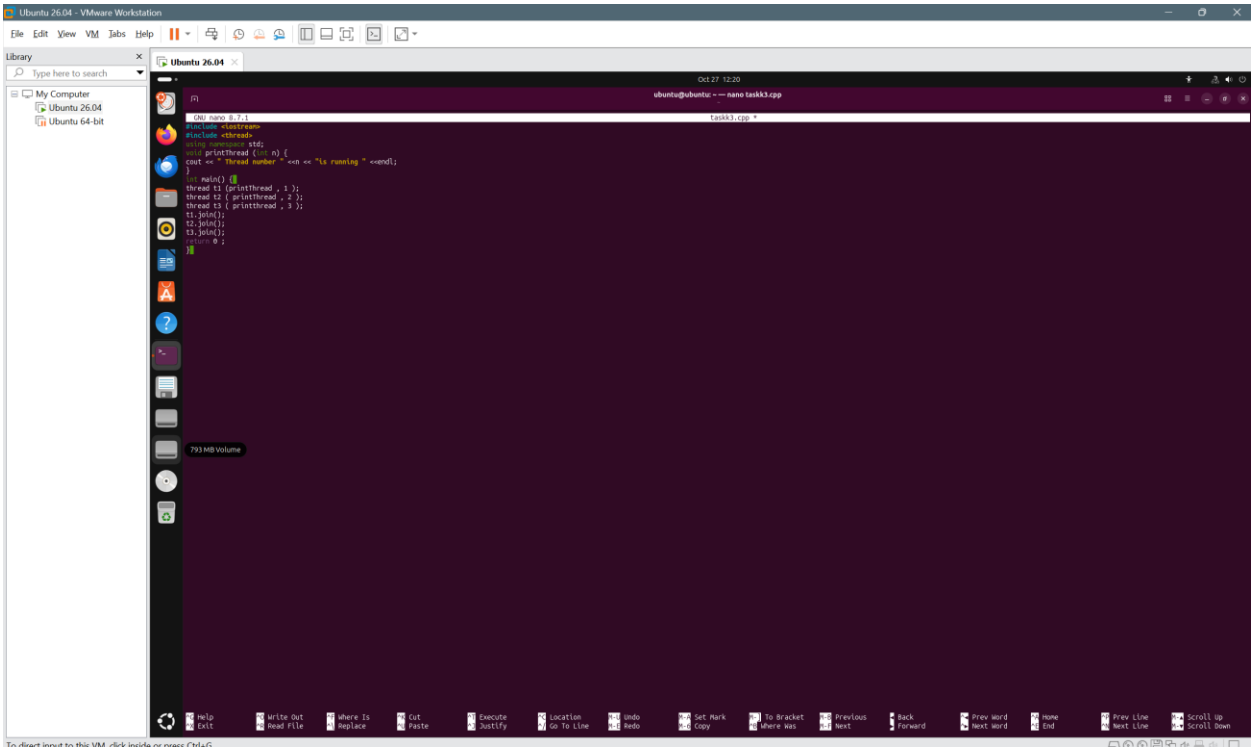

Ubuntu 26.04 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Ubuntu 26.04
Ubuntu 64-bit
Ubuntu 26.04
Oct 27 12:12
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap task1 task1.cpp
ubuntu@ubuntu:~$ cd downloads
bash: cd: downloads: No such file or directory
ubuntu@ubuntu:~$ touch file.txt
ubuntu@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos file.txt snap task1 task1.cpp
ubuntu@ubuntu:~$ oscode.txt
oscode.txt: command not found
ubuntu@ubuntu:~$ open file.txt
ubuntu@ubuntu:~$ cat file.txt
maha
Aman
Musfirah
Noor
Aman
ubuntu@ubuntu:~$ cat file.txt | grep 'Maha'
ubuntu@ubuntu:~$ cat file.txt | grep 'maha'
maha
ubuntu@ubuntu:~$ nano task1.cpp
ubuntu@ubuntu:~$ nano taskk1.cpp
ubuntu@ubuntu:~$ g++ taskk1.cpp -o taskk1 -pthread
taskk1.cpp: In function 'int main()':
taskk1.cpp:9:12: error: 'threadFuntion' was not declared in this scope; did you mean 'threadFunction'?
9 | thread t (threadFuntion) ;
| ^~~~~~~~~~~~
| threadFunction
ubuntu@ubuntu:~$ nano taskk1.cpp
ubuntu@ubuntu:~$ g++ taskk1.cpp -o taskk1 -pthread
ubuntu@ubuntu:~$ ./taskk1.cpp
bash: ./taskk1.cpp: Permission denied
ubuntu@ubuntu:~$ ./taskk1
hello from the thread
main thread finished
ubuntu@ubuntu:~$ nano taskk1.cpp
ubuntu@ubuntu:~$ nano taskk2.cpp
ubuntu@ubuntu:~$ g++ taskk2.cpp -o taskk2 -pthread
taskk2.cpp:1:10: fatal error: iostream: No such file or directory
1 | #include < iostream>
| ^~~~~~~~~~~~
compilation terminated.
ubuntu@ubuntu:~$ nano taskk2.cpp
ubuntu@ubuntu:~$ g++ taskk2.cpp -o taskk2 -pthread
ubuntu@ubuntu:~$ ./taskk2
Sum = 30
Sum = 40
ubuntu@ubuntu:~$

Task 4: Running Multiple Threads Together

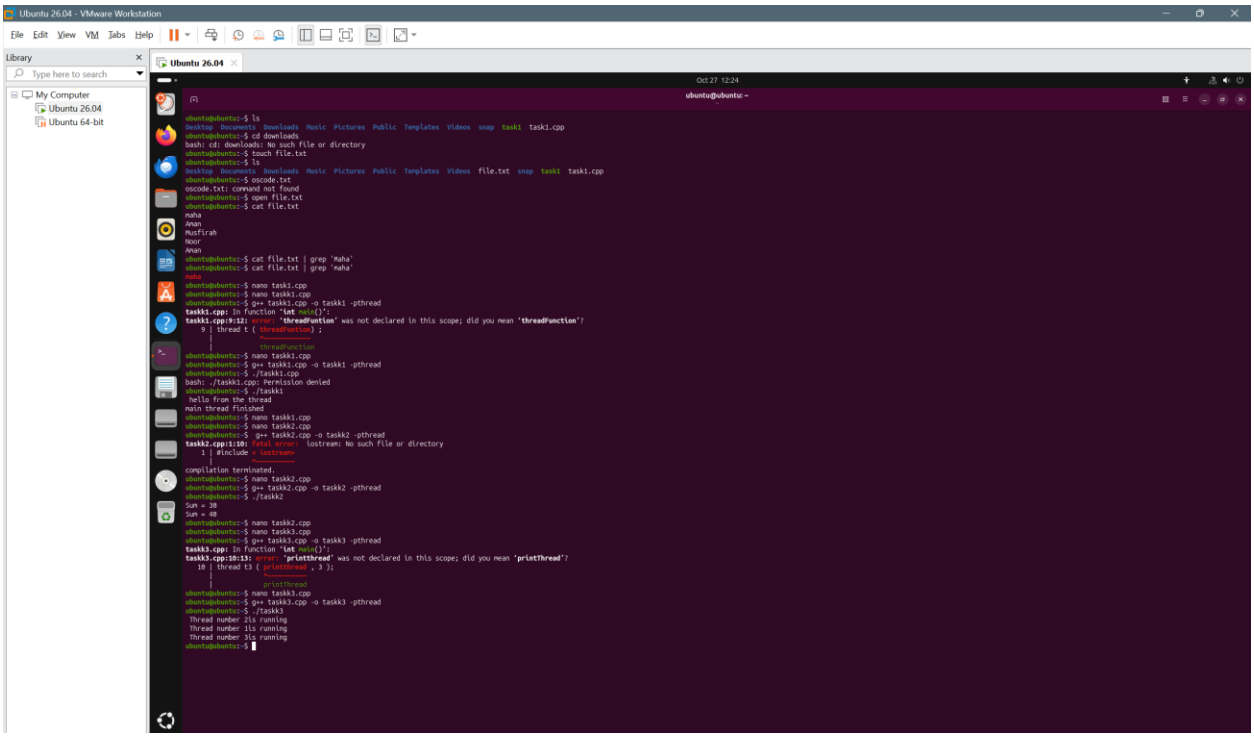
Explanation: This program creates multiple threads that run the same function with different arguments. The function printThread(int n) prints "Thread number n is running". Three threads (t1, t2, t3) are created with the values 1, 2 and 3, and then each one is joined so that main waits for all of them to finish before the program ends.

In the output, the threads did not print in the order they were created (Thread 2 appeared before Thread 1). This happens because threads run concurrently and the operating system scheduler decides which one runs first, so the order is not guaranteed. A compile error caused by a misspelled function name (printthread instead of printThread) was fixed before the final run.

Code :



Output:


Ubuntu 26.04 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Ubuntu 26.04
Ubuntu 64-bit
Ubuntu 26.04
Oct 27 12:24
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap task1 task1.cpp
ubuntu@ubuntu:~$ cd downloads
bash: cd: downloads: No such file or directory
ubuntu@ubuntu:~$ touch file.txt
ubuntu@ubuntu:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos file.txt snap task1 task1.cpp
ubuntu@ubuntu:~$ oscode.txt
oscode.txt: command not found
ubuntu@ubuntu:~$ open file.txt
ubuntu@ubuntu:~$ cat file.txt
maha
Anan
Musfirah
Noor
Anan
ubuntu@ubuntu:~$ cat file.txt | grep 'Maha'
ubuntu@ubuntu:~$ cat file.txt | grep 'maha'
maha
ubuntu@ubuntu:~$ nano task1.cpp
ubuntu@ubuntu:~$ nano task1.cpp
ubuntu@ubuntu:~$ g++ task1.cpp -o task1 -pthread
task1.cpp: In function 'int main()':
task1.cpp:9:12: error: 'threadFuntion' was not declared in this scope; did you mean 'threadFunction'?
9 | thread t { threadFuntion} ;
| ^~~~~~~~~~~~
| threadFunction
ubuntu@ubuntu:~$ nano task1.cpp
ubuntu@ubuntu:~$ g++ task1.cpp -o task1 -pthread
ubuntu@ubuntu:~$ ./task1.cpp
bash: ./task1.cpp: Permission denied
ubuntu@ubuntu:~$ ./task1
hello from the thread
main thread finished
ubuntu@ubuntu:~$ nano task1.cpp
ubuntu@ubuntu:~$ nano task2.cpp
ubuntu@ubuntu:~$ g++ task2.cpp -o task2 -pthread
task2.cpp:1:10: fatal error: iostream: No such file or directory
1 | #include <iostream>
| ^~~~~~~~~~
compilation terminated.
ubuntu@ubuntu:~$ nano task2.cpp
ubuntu@ubuntu:~$ g++ task2.cpp -o task2 -pthread
ubuntu@ubuntu:~$ ./task2
Sum = 30
Sum = 40
ubuntu@ubuntu:~$ nano task2.cpp
ubuntu@ubuntu:~$ nano task3.cpp
ubuntu@ubuntu:~$ g++ task3.cpp -o task3 -pthread
task3.cpp: In function 'int main()':
task3.cpp:10:13: error: 'printthread' was not declared in this scope; did you mean 'printThread'?
10 | thread t3 { printthread, 3 };
| ^~~~~~~~~~~
| printThread
ubuntu@ubuntu:~$ nano task3.cpp
ubuntu@ubuntu:~$ g++ task3.cpp -o task3 -pthread
ubuntu@ubuntu:~$ ./task3
Thread number 2is running
Thread number 1is running
Thread number 3is running
ubuntu@ubuntu:~$